

NAME: _____

PERIOD: _____

DATE: _____

When Two Proportions Are Not Independent

AP Statistics · Unit 3 · Topic 3.9 · B: 2026-12-10 · E: 2027-01-20

[STAR] TODAY'S PROBLEM

A library studies self-checkout before and after a redesign. Records give $p_B = 0.42$ among 4,000 before visits and $p_A = 0.54$ among 5,000 after visits.

Plan I: Draw an SRS of 120 before visits and a separate SRS of 150 after visits; do not track cardholders.

Plan P: Measure the same 120 cardholders before and after.

Sora: Separate selection warrants Plan I's independent model; 0.0350 means a difference at most 0.01 is unusual.

Mateo: Pairing controls person differences, so Plan P may help. Because both proportions use 120 people, Plan I's model applies to P too.

Sampling plans

Plan	Before	After	Repeated
I	SRS 120	SRS 150	no
P	120	same 120	yes

FIRST TAKE — before the video (5 min)

Which parts of Sora's and Mateo's reasoning are warranted? Cite one design detail.

[BOOK] THE VARIANCE FORMULA REQUIRES INDEPENDENT SAMPLES (keep on the page)

For independent samples, $\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$ and $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{p_1(1 - p_1)/n_1 + p_2(1 - p_2)/n_2}$ when each sample is below 10% of its population. Approximate normality requires all four expected counts to be at least 10. Repeated measurements on the same people are paired, not independent; their association belongs in the variability.

Finish after the video:

DOK 1 (a) For after minus before, identify the parameter and statistic and state Plan I's sampling-distribution mean.

DOK 2 (b) For Plan I, check both 10 percent conditions and all four counts. Find the standard deviation and approximate $P(\hat{p}_A - \hat{p}_B \leq 0.01)$; interpret it.

DOK 3 (c) Adjudicate the claims. Use design and conditions to place the independent model and assess Plan P. Explain why equal sample sizes are not independence, how transferring 0.0607 and 0.0350 could mislead, and what paired analysis must retain.

Frame: Plan _____ permits the independent model because _____, and its four counts are _____.

Frame: For Plan P, the missing _____ could make 0.0350 _____; a valid analysis must keep _____.

EXIT — turn this in

One thing the video changed about my first take: _____